

# Mazur's Program B, after Furio–Lombardo

David Zureick-Brown (Amherst College)  
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ChaBONNty, at the Max Plank Institute for Mathematics

July 2, 2026

Slides available at <https://dmzb.github.io/>

## Summary of what's new:

- We solve the final two 7-adic cases of program B (in progress)
- using Furio–Lombardo's recent work, and
- a “variant” of Kummer's proof of FLT for regular primes

# Condensed Pseudo-Perfectoid Higher $\infty$ -Stacks and a Derived Hyper-Monoidal Liquid Arithmetic Bhargava–Gauss–Langlands–Scholze–Rota–Witt Correspondence via $\delta$ -modularity of $\ell$ -adic forms on the Rigid Interuniversal Log-Crystalline Site of Non-Teichmüller Motives: A Gentle Introduction<sup>1</sup>

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$$\begin{array}{c}
 \begin{array}{ccc}
 \tilde{X}_A & \xleftrightarrow{\alpha_u} & \tilde{X}_A \\
 \downarrow \psi_B & \searrow f & \downarrow \psi_B \\
 \Pi_1(G) & \xrightarrow{\quad} & X_B
 \end{array}
 \end{array}
 \quad
 \begin{array}{c}
 D^+(R_{\infty}(1/p) \otimes \mathcal{L} \mathcal{L}^{\dagger}(S) = R\Gamma(\widehat{Y}_B, \widehat{\mathcal{D}}_{LG} \widehat{T}) \longrightarrow \int_0^{\infty} \frac{M}{\mathcal{T}_E n} \\
 - |x|_{dd} < e. \implies R\Gamma(\widehat{Y}'_B, \widehat{\mathcal{D}}_{LT}) \approx R\mathrm{Hom}(S, T) \Big|_{\mathcal{O}_B}^{\mathcal{O}_B} = \frac{M}{\mathcal{T}_E n} \\
 \sum_z i\{n \operatorname{Ext}_1^z(M, N) = [-\mathcal{T}_C \mathcal{O} \cdot \operatorname{Gal}(L/M) \triangleleft \operatorname{GL}_2(\dot{E})^{L+} \} \\
 \lim_{\alpha \infty} = E_k^{\tau} \quad \tau_n = \operatorname{Tr}_{\mathcal{O}}(\Phi^{\mathcal{C}_1} V_1) \rightarrow 0 \rightarrow F_E \rightarrow C^l \rightarrow 0 \quad \longleftarrow \circ \longrightarrow \quad \longleftarrow \frac{\widehat{\mathcal{T}}_C + H}{\mathcal{O}_{\mathcal{A}} + H} = \\
 \xrightarrow{x \mapsto x + \dots} |D_j^a| > \operatorname{RH}^{\Delta}(\widehat{V}_{\psi}, \overline{V}) - H^{\frac{u_a}{\theta}}(X, \overline{Z}_p) = (\widehat{T}_p)^{\frac{n}{2}} \Big\| \operatorname{Mod}_{\Delta} \Leftrightarrow \widehat{D}F\text{-Isoc} \circ.
 \end{array}$$

<sup>1</sup>but with Full Proofs

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# Galois Representations

$$\begin{aligned}\mathbb{Q} &\subset K \subset \overline{\mathbb{Q}} \\ G_K &:= \text{Aut}(\overline{K}/K) \\ E[n](\overline{K}) &\cong (\mathbb{Z}/n\mathbb{Z})^2\end{aligned}$$

$$\begin{aligned}\rho_{E,n}: G_K &\rightarrow \text{Aut } E[n] \cong \text{GL}_2(\mathbb{Z}/n\mathbb{Z}) \\ \rho_{E,\ell^\infty}: G_K &\rightarrow \text{GL}_2(\mathbb{Z}_\ell) = \varprojlim_n \text{GL}_2(\mathbb{Z}/\ell^n\mathbb{Z}) \\ \rho_E: G_K &\rightarrow \text{GL}_2(\widehat{\mathbb{Z}}) = \varprojlim_n \text{GL}_2(\mathbb{Z}/n\mathbb{Z})\end{aligned}$$

# Image of Galois

$$\rho_{E,n}: G_{\mathbb{Q}} \twoheadrightarrow H(n) \hookrightarrow \mathrm{GL}_2(\mathbb{Z}/n\mathbb{Z})$$

$$\left\{ \begin{array}{c} \overline{\mathbb{Q}} \\ | \\ \overline{\mathbb{Q}}^{\ker \rho_{E,n}} = \mathbb{Q}(E[n]) \\ | \\ \mathbb{Q} \end{array} \right\} H(n)$$

Problem (Mazur's "program B")

*Classify all possibilities for  $H(n)$ .*

# Mazur's Program B

As presented at Modular functions in one variable V in Bonn

Theorem 1 also fits into a general program:

B. Given a number field  $K$  and a subgroup  $H$  of  $GL_2(\hat{\mathbb{Z}}) = \prod_p GL_2(\mathbb{Z}_p)$  classify  
all elliptic curves  $E/K$  whose associated Galois representation on torsion points  
maps  $Gal(\bar{K}/K)$  into  $H \subset GL_2(\hat{\mathbb{Z}})$ .

Mazur - Rational points on modular curves (1977)

## Example - torsion on an elliptic curve

If  $E$  has a  $K$ -rational **torsion point**  $P \in E(K)[n]$  (of exact order  $n$ ) then:

$$H(n) \subset \begin{pmatrix} 1 & * \\ 0 & * \end{pmatrix}$$

since for  $\sigma \in G_K$  and  $Q \in E(\overline{K})[n]$  such that  $E(\overline{K})[n] \cong \langle P, Q \rangle$ ,

$$\begin{aligned}\sigma(P) &= P \\ \sigma(Q) &= a_\sigma P + b_\sigma Q\end{aligned}$$

## Example - Isogenies

If  $E$  has a  $K$ -rational, **cyclic isogeny**  $\phi: E \rightarrow E'$  with  $\ker \phi = \langle P \rangle$  then:

$$H(n) \subset \begin{pmatrix} * & * \\ 0 & * \end{pmatrix}$$

since for  $\sigma \in G_K$  and  $Q \in E(\overline{K})[n]$  such that  $E(\overline{K})[n] \cong \langle P, Q \rangle$ ,

$$\begin{aligned} \sigma(P) &= a_\sigma P \\ \sigma(Q) &= b_\sigma P + c_\sigma Q \end{aligned}$$



## Example - other maximal subgroups

$\mathbb{F}_{p^2}^*$  acts on  $\mathbb{F}_{p^2} \cong \mathbb{F}_p \times \mathbb{F}_p$

**Normalizer of a non-split Cartan:**

$$C_{\text{ns}} = \text{im} \left( \mathbb{F}_{p^2}^* \rightarrow \text{GL}_2(\mathbb{F}_p) \right) \subset N_{\text{ns}}$$

$H(n) \subset N_{\text{ns}}$  and  $H(n) \not\subset C_{\text{ns}}$  iff

$E$  admits a “necklace” (Rebolledo, Wuthrich)

# Modular curves

## Definition

- $X(N)(K) := \{(E/K, P, Q) : E[N] = \langle P, Q \rangle\} \cup \{\text{cusps}\}$
- $X(N)(K) \ni (E/K, P, Q) \Leftrightarrow \rho_{E,N}(G_K) = \{I\}$

Let  $\Gamma(N) \subset H \subset \text{GL}_2(\widehat{\mathbb{Z}})$ . The minimal such  $N$  is the **level** of  $H$ .

## Definition

$X_H := X(N)/H(N)$  (where  $H(N)$  is the image of  $H$  in  $\text{GL}_2(\mathbb{Z}/N\mathbb{Z})$ )

$$X_H(K) \ni (E/K, \iota) \Leftrightarrow \rho_{E,N}(G_K) \subset H(N)$$

## Stacky disclaimer

This is only true up to twist; there are some subtleties if

- 1  $j(E) \in \{0, 12^3\}$  (plus some minor group theoretic conditions), or
- 2 if  $-I \in H$ .

# Rational Points on modular curves

## Mazur's program B

Compute  $X_H(\mathbb{Q})$  for all  $H$ .

## Remark

- Sometimes  $X_H \cong \mathbb{P}^1$  or elliptic with rank  $X_H(\mathbb{Q}) > 0$
- Some  $X_H$  have **exceptional** points  
(i.e.  $X_H(\mathbb{Q})$  is finite and  $E$  is non-cuspidal and non-CM)
- Can compute  $g(X_H)$  group theoretically (via Riemann–Hurwitz).

## Fact

$$g(X_H), \gamma(X_H) \rightarrow \infty \text{ as } [\mathrm{GL}_2(\widehat{\mathbb{Z}}) : H] \rightarrow \infty.$$

Let  $E$  be an elliptic curve over  $\mathbb{Q}$  without CM

Uniformity conjecture / Question of Serre

For  $\ell > 37$ ,  $\rho_{E,\ell}$  is surjective.

Conjecture (Zywina)

$$\left[ \mathrm{GL}_2(\widehat{\mathbb{Z}}) : \rho_E(G_{\mathbb{Q}}) \right] \leq 2736.$$

Zywina: the indices occurring infinitely often divide

220, 336, 360, 504, 864, 1152, 1200, 1296 or 1536

(modulo conjectures)

Remark

$j(E) \in \{-7 \cdot 11^3, -7 \cdot 137^3 \cdot 2083^3\}$  *does occur finitely often*

## Recall

A point  $(E, \iota) \in X_H(K)$  is **exceptional** if  $X_H(K)$  is finite and  $\text{End } E = \mathbb{Z}$ .

Let  $\ell$  prime,  $E/\mathbb{Q}$  be a non-CM elliptic curve, and  $H = \rho_{E, \ell^\infty}(G_{\mathbb{Q}})$ .

## Theorem (Rouse–Sutherland–ZB 2021)

*Exactly one of the following is true:*

- ①  $X_H(\mathbb{Q})$  is **infinite** and  $H$  is listed in (Sutherland–Zywina 2017);
- ②  $X_H$  has a rational **exceptional** point listed in Table 1;
- ③  $H \leq N_{\text{ns}}(3^3), N_{\text{ns}}(5^2), N_{\text{ns}}(7^2), N_{\text{ns}}(11^2)$ , or  $N_{\text{ns}}(\ell)$  for some  $\ell > 13$ ; or
- ④  $H \leq G_{\text{sp}}^\#(7^2), G_{\text{ns}}^\#(7^2)$  (49.179.9.1 or 49.196.9.1).

## Conjecture

Cases (3) and (4) **never occur**.

If they do, the exceptional points have **extraordinarily** large heights (e.g.  $10^{10^{200}}$  for  $X_{\text{ns}}^+(11^2)(\mathbb{Q})$ ).

| label       | level          | notes                              | $j$ -invariants/models of exceptional points                                                                                                                                                                     |
|-------------|----------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16.64.2.1   | 2 <sup>4</sup> | $N_{\text{ns}}(16)$                | $-2^{18} \cdot 3 \cdot 5^3 \cdot 13^3 \cdot 41^3 \cdot 107^3 / 17^{16}$<br>$-2^{21} \cdot 3^3 \cdot 5^3 \cdot 7 \cdot 13^3 \cdot 23^3 \cdot 41^3 \cdot 179^3 \cdot 409^3 / 79^{16}$                              |
| 16.96.3.335 | 2 <sup>4</sup> | $H(4) \subsetneq N_{\text{sp}}(4)$ | $257^3 / 2^8$                                                                                                                                                                                                    |
| 16.96.3.343 | 2 <sup>4</sup> | $H(4) \subsetneq N_{\text{sp}}(4)$ | $17^3 \cdot 241^3 / 2^4$                                                                                                                                                                                         |
| 16.96.3.346 | 2 <sup>4</sup> | $H(4) \subsetneq N_{\text{sp}}(4)$ | $2^4 \cdot 17^3$                                                                                                                                                                                                 |
| 16.96.3.338 | 2 <sup>4</sup> | $H(4) \subsetneq N_{\text{sp}}(4)$ | $2^{11}$                                                                                                                                                                                                         |
| 32.96.3.230 | 2 <sup>5</sup> | $H(4) \subsetneq N_{\text{sp}}(4)$ | $-3^3 \cdot 5^3 \cdot 47^3 \cdot 1217^3 / (2^8 \cdot 31^8)$                                                                                                                                                      |
| 32.96.3.82  | 2 <sup>5</sup> | $H(8) \subsetneq N_{\text{sp}}(8)$ | $3^3 \cdot 5^6 \cdot 13^3 \cdot 23^3 \cdot 41^3 / (2^{16} \cdot 31^4)$                                                                                                                                           |
| 25.50.2.1   | 5 <sup>2</sup> | $H(5) = N_{\text{ns}}(5)$          | $2^4 \cdot 3^2 \cdot 5^7 \cdot 23^3$                                                                                                                                                                             |
| 25.75.2.1   | 5 <sup>2</sup> | $H(5) = N_{\text{sp}}(5)$          | $2^{12} \cdot 3^3 \cdot 5^7 \cdot 29^3 / 7^5$                                                                                                                                                                    |
| 7.56.1.2    | 7              | $\subsetneq N_{\text{ns}}(7)$      | $3^3 \cdot 5 \cdot 7^5 / 2^7$                                                                                                                                                                                    |
| 7.112.1.2   | 7              | $-I \notin H$                      | $y^2 + xy + y = x^3 - x^2 - 2680x - 50053$<br>$y^2 + xy + y = x^3 - x^2 - 131305x + 17430697$                                                                                                                    |
| 11.60.1.3   | 11             | $\subsetneq B(11)$                 | $-11 \cdot 131^3$                                                                                                                                                                                                |
| 11.120.1.8  | 11             | $-I \notin H$                      | $y^2 + xy + y = x^3 + x^2 - 30x - 76$                                                                                                                                                                            |
| 11.120.1.9  | 11             | $-I \notin H$                      | $y^2 + xy = x^3 + x^2 - 2x - 7$                                                                                                                                                                                  |
| 11.60.1.4   | 11             | $\subsetneq B(11)$                 | $-11^2$                                                                                                                                                                                                          |
| 11.120.1.3  | 11             | $-I \notin H$                      | $y^2 + xy = x^3 + x^2 - 3632x + 82757$                                                                                                                                                                           |
| 11.120.1.4  | 11             | $-I \notin H$                      | $y^2 + xy + y = x^3 + x^2 - 305x + 7888$                                                                                                                                                                         |
| 13.91.3.2   | 13             | $S_4(13)$                          | $2^4 \cdot 5 \cdot 13^4 \cdot 17^3 / 3^{13}, -2^{12} \cdot 5^3 \cdot 11 \cdot 13^4 / 3^{13}$<br>$2^{18} \cdot 3^3 \cdot 13^4 \cdot 127^3 \cdot 139^3 \cdot 157^3 \cdot 283^3 \cdot 929 / (5^{13} \cdot 61^{13})$ |
| 17.72.1.2   | 17             | $\subsetneq B(17)$                 | $-17 \cdot 373^3 / 2^{17}$                                                                                                                                                                                       |
| 17.72.1.4   | 17             | $\subsetneq B(17)$                 | $-17^2 \cdot 101^3 / 2$                                                                                                                                                                                          |
| 37.114.4.1  | 37             | $\subsetneq B(37)$                 | $-7 \cdot 11^3$                                                                                                                                                                                                  |
| 37.114.4.2  | 37             | $\subsetneq B(37)$                 | $-7 \cdot 137^3 \cdot 2083^3$                                                                                                                                                                                    |

Table 1. All known exceptional groups,  $j$ -invariants, and points of prime power level.

# Approach

Compute the set of **arithmetically maximal** subgroups

$H$  is arithmetically maximal if

- $X_H(\mathbb{Q})$  is finite and
- for every  $G$  such that  $H \leq G$ ,  $X_G(\mathbb{Q})$  is infinite.

For such  $H$  **compute equations** for  $X_H$  and  $j_H: X_H \rightarrow X(1)$

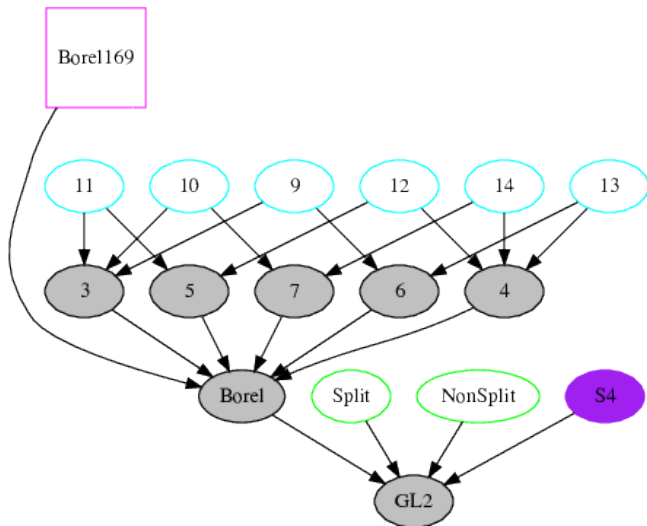
See Rouse's **VaNTAGe talk** for details and examples,

Assaf's **recent paper** and

Zywina's **BIRS talk** for newer efficient approaches.

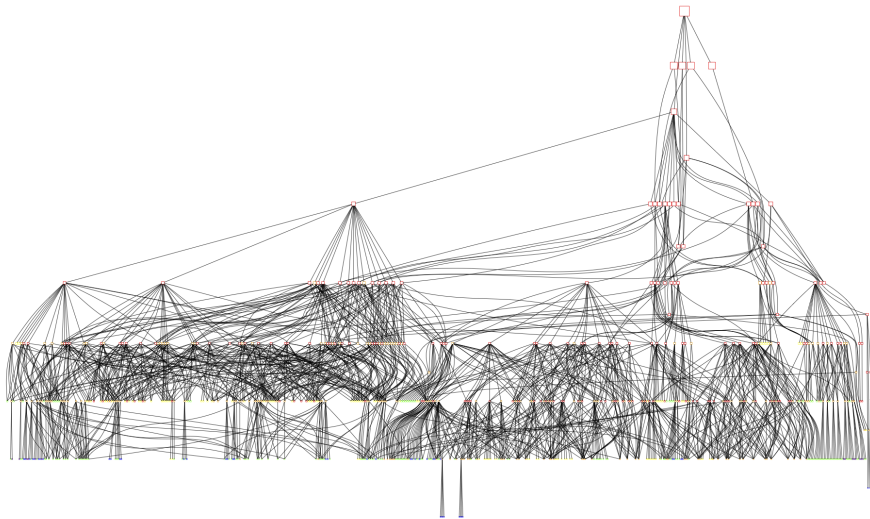
For such  $H$  with  $-I \in H$  **determine the rational points** in  $X_H(\mathbb{Q})$

# Subgroups of $GL_2(\mathbb{Z}_{13})$





# Subgroups of $GL_2(\mathbb{Z}_2)$



## Explicit methods: highlight reel

- Local methods
  - Chabauty and Elliptic Chabauty
  - Mordell–Weil sieve
  - étale descent
  - Pryms
  - *Equationless étale descent via group theory*
  - *New techniques for computing  $\text{Aut } C$*
- 
- *Nonabelian Chabauty*
  - **“Equationless” local methods** and **Mordell–Weil sieve**
  - **Greenberg Transforms** (and big computations)
  - **Novel variants of existing techniques** (e.g., Atkin–Lehner sieve)
  - **Modularity of isogeny factors of  $J_H$**  (w/ Voight)

# UNSOLVED *mysteries*

Arithmetically maximal level  $\ell^n$  groups with  $\ell \leq 13$  with  $X_H(\mathbb{Q})$  **unknown**.

| label          | level  | group                     | genus |
|----------------|--------|---------------------------|-------|
| 27.243.12.1    | $3^3$  | $N_{\text{ns}}(3^3)$      | 12    |
| 25.250.14.1    | $5^2$  | $N_{\text{ns}}(5^2)$      | 14    |
| 49.1029.69.1   | $7^2$  | $N_{\text{ns}}(7^2)$      | 69    |
| 49.147.9.1     | $7^2$  | $G_{\text{ns}}^{\#}(7^2)$ | 9     |
| 49.196.9.1     | $7^2$  | $G_{\text{sp}}^{\#}(7^2)$ | 9     |
| 121.6655.511.1 | $11^2$ | $N_{\text{ns}}(11^2)$     | 511   |

Each has **rank = genus**, **rational CM points**, **no rational cusps**, and **no known exceptional points**.



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| <del>27.243.12.1</del> <sup>2</sup>  | $3^3$  | $N_{\text{ns}}(3^3)$      | 12    |
| 25.250.14.1                          | $5^2$  | $N_{\text{ns}}(5^2)$      | 14    |
| <del>49.1029.69.1</del> <sup>3</sup> | $7^2$  | $N_{\text{ns}}(7^2)$      | 69    |
| 49.147.9.1                           | $7^2$  | $G_{\text{ns}}^{\#}(7^2)$ | 9     |
| 49.196.9.1                           | $7^2$  | $G_{\text{sp}}^{\#}(7^2)$ | 9     |
| 121.6655.511.1                       | $11^2$ | $N_{\text{ns}}(11^2)$     | 511   |

Each has **rank = genus**, **rational CM points**, **no rational cusps**, and **no known exceptional points**.

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<sup>2</sup>Balakrishnan–Betts–Hast–Jha–Müller

<sup>3</sup>Furio–Lombardo

# Stacks and ramification

## Some related facts:

- The map  $X_{\text{ns}}(\ell^n) \rightarrow X(1)$  is ramified with degree  $\ell^n$  at each cusp.
- Thus  $X_{\text{ns}}(\ell^n) \rightarrow X_{\text{ns}}(\ell^{n-i})$  is ramified with degree  $\ell^i$  at each cusp.
- The denominator of  $j(E)$  is a perfect  $\ell^n$ th power (for  $E$  a point on  $X_{\text{ns}}(\ell^n)$  and  $\ell$  odd).
- The denominator of the  $j$ -map  $X_{\text{ns}}(\ell^n) \rightarrow X(1)$  is “often” a perfect  $\ell^n$ th power.

## Denominator of $j$ -map

$$j_{\text{ns}}: X_{\text{ns}}(7) \cong \mathbb{P}^1 \rightarrow X(1)$$

is given by

$$j(E) = j_{\text{ns}}(t) = \frac{g(x, y)^3}{f(x, y)^7}$$

The denominator of  $j_{\text{ns}}: X_{\text{ns}}(7^2) \rightarrow X(1)$  is a  $7^2$ th power.

So if  $j(E)$  is the  $j$ -invariant of an exceptional point on  $X_{\text{ns}}(7^2)$ ,  
then  $f(x, y)^7$  is a  $7^2$ th power (up to a resultant),  
i.e.,  $f(x, y)$  is a 7th power.  
(up to resultants)

**Furio–Lombardo:** Let  $E/\mathbb{Q}$  be an elliptic curve without CM.

If  $\text{Im } \rho_{E,49} \subseteq C_{\text{ns}}^+(49)$ , there exist  $k \in \{1, 8\}$  and pairwise coprime integers  $x, y, z \in \mathbb{Z}$  such that

$$x^3 - 7x^2y + 7xy^2 + 7y^3 = k \cdot z^7 \quad \text{and} \quad j(E) = j_{\text{ns}} \left( \frac{x}{y} \right).$$

If  $\text{Im } \rho_{E,49} \subseteq G_{\text{ns}}^{\#}(49)$ , there exist  $k \in \{1, 8\}$  and pairwise coprime integers  $x, y, z \in \mathbb{Z}$  such that

$$x^3 - 7x^2y + 7xy^2 + 7y^3 = 7k \cdot z^7 \quad \text{and} \quad j(E) = j_{\text{ns}} \left( \frac{x}{y} \right).$$

If  $\text{Im } \rho_{E,49} \subseteq G_{\text{sp}}^{\#}(49)$ , there exist  $k \in \{1, 7\}$  and pairwise coprime integers  $x, y, z, w \in \mathbb{Z}$  such that  $y = w^7$  and

$$x^3 - 4x^2y + 3xy^2 + y^3 = k \cdot z^7 \quad \text{and} \quad j(E) = j_{\text{sp}} \left( \frac{x}{y} \right).$$

Darmon–Granville: Faltings' theorem is true for stacky curves.

Certain weighted homogenous surfaces have finitely many primitive integral solutions, e.g.,

- $x^3 + y^5 = z^7$ ,
- $x^3 - 7x^2y + 7xy^2 + 7y^3 = k \cdot z^7$ ,

since their stack quotients are hyperbolic.

Bennett–Dahmen: Generalized superelliptic equations

If  $F(x, y)$  is a homogenous binary cubic, then

$$4H(x, y)^3 + G(x, y)^2 = -27\Delta_F F(x, y)^2.$$

Thus

$$-27\Delta(x^3 - 7x^2y + 7xy^2 + 7y^3)^2 = 4H(x, y)^3 + G(x, y)^2 = -27(k \cdot z^7)^2$$



## Theorem (Furio–Lombardo)

The  $(\star)$ -solutions of the equation

$$a^2 + 28b^3 = 27c^7$$

are precisely

$$(\pm 1, -1, -1), \quad (\pm 27, -3, -1), \quad (\pm 2521, -61, -1).$$

## The Poonen–Schaefer–Stoll playbook

- $(a, b, c) \mapsto E_{a,b,c}$
- $\exists E$  with  $N_E \nmid N$  such that  $E_{a,b,c}[7] \cong E[7]$
- This corresponds to a point  $(E_{a,b,c}[7], \iota) \in X_E(7)(\mathbb{Q})$ .
- $X_E(7)$  is a twist of the Klein quartic, a genus 3 plane quartic.

$$x^4 + 3x^3y - 3x^2yz - 3x^2z^2 + 6xy^3 - 6xy^2z + 3xyz^2 - 2xz^3 + 4y^4 + 2y^3z - 5yz^3$$

- One can do Chabauty and more with this finite list of curves.

Theorem (ZB–Santiago Arango-Piñeros, **in progress**)

$X_{\text{ns}}^{\#}(49)$  and  $X_{\text{sp}}^{\#}(49)$  *have no exceptional points*

# Kummer: Fermat's Last Theorem for Regular primes

## Theorem

*If  $(x, z, w) \in \mathbb{Z}^3$  is a solution to  $x^7 + z^7 = w^7$ , then 7 divides  $xzw$ .*

**Idea:** factor over  $L = \mathbb{Q}(\zeta_7)$ , and use these facts about  $L$  and  $K = \mathbb{Q}(\zeta_7 + \zeta_7^{-1}) \subset \mathbb{R}$ :

- 1  $\mathbb{Q}(\zeta_7)$  has class number 1
- 2  $\mathcal{O}_L^\times = \mu_L \mathcal{O}_K^\times$
- 3 for all  $\beta \in L$ ,  $\beta^7 = b \pmod{7\mathcal{O}_L}$  for some  $b \in \mathbb{Z}$ .

Let  $L = \mathbb{Q}(\zeta_7)$  and  $K = \mathbb{Q}(\zeta_7 + \zeta_7^{-1}) \subset \mathbb{R}$ . Then

$$x^7 + z^7 = \prod (x - \zeta_7^i z) = w^7$$

Since  $\text{cl}(L) = 1$  and  $\mathcal{O}_L^\times = \mu_L \mathcal{O}_K^\times$

$$x - \zeta_7 z = (\zeta_7^r u) \beta^7, \quad \text{for } u \in \mathcal{O}_K^\times, \quad \beta \in \mathcal{O}_L.$$

Elementary:  $\beta^7 \equiv b \pmod{7\mathcal{O}_L}$  for some  $b \in \mathbb{Z}$ . Thus

$$x - \zeta_7 z = \zeta_7^r u b$$

Applying complex conjugation gives

$$x - \zeta_7^{-1} z = \zeta_7^{-r} u b$$

Combining

$$\zeta_7^r (x - \zeta_7^{-1} z) \equiv \zeta_7^{-r} (x - \zeta_7 z) \pmod{7\mathcal{O}_L}$$

i.e.,

$$x \cdot 1 - z \cdot \zeta_7 + z \cdot \zeta_7^{2r-1} - x \cdot \zeta_7^{2r} \equiv 0 \pmod{7\mathcal{O}_L}.$$

If the elements  $1, \zeta_7, \zeta_7^{2r-1}, \zeta_7^{2r}$  are all distinct, then they are part of a  $\mathbb{Z}$  basis for  $\mathcal{O}_L = \mathbb{Z}[\zeta_7]$ , and so implies that  $7 \mid x$  and  $7 \mid z$ .

## A “variant” of Kummer’s proof

Suppose that we want to solve

$$x^3 - 4x^2z + 3xz^2 + z^3 = w^7.$$

Kummer’s approach gives  $u \in \mathcal{O}_L^*/(\mathcal{O}_L^*)^7$  such that

$$x - z\alpha = u\beta^7$$

Extend  $\{1, \alpha\}$  to a basis. ( $\alpha^2$  will do fine.) Write

$$\beta = a + b\alpha + c\alpha^2$$

where  $a, b, c$  are variables. Then

$$x - z\alpha = \beta^7 = (a + b\alpha + c\alpha^2)^7 = f_0(a, b, c) + f_1(a, b, c)\alpha + f_2(a, b, c)\alpha^2$$

for some homogenous degree 7 polynomials  $f_i$ . In particular,

$$x = f_0(a, b, c)$$

$$-z = f_1(a, b, c)$$

$$0 = f_2(a, b, c).$$

This last equation  $f_2(a, b, c)$  defines a smooth genus 15 curve  $C$ .

$$x - z\alpha = \beta^7 = (a + b\alpha + c\alpha^2)^7 = f_0(a, b, c) + f_1(a, b, c)\alpha + f_2(a, b, c)\alpha^2$$

$f_2(a, b, c)$  defines a smooth genus 15 curve  $C$ .

Over  $K = \mathbb{Q}(\zeta_7)$  we have computable maps

$$C \rightarrow D \rightarrow E$$

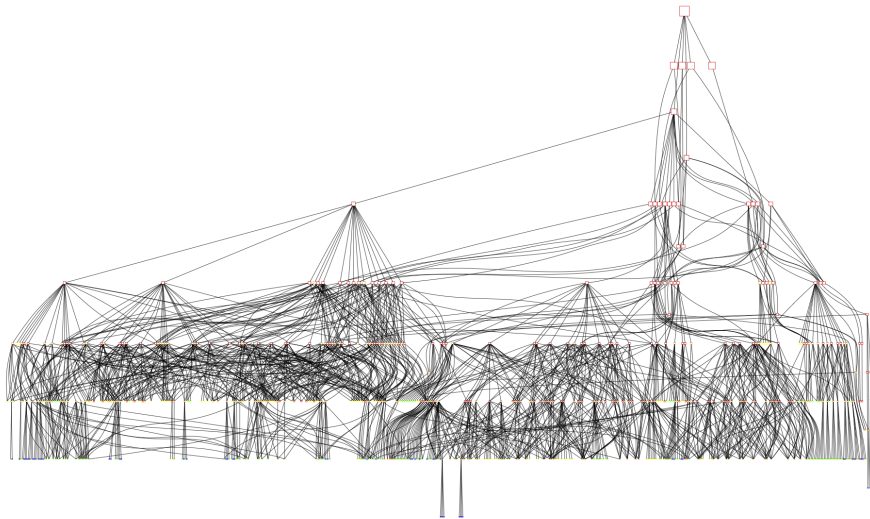
where  $D$  is a genus 3 curve and  $E$  is an elliptic curve with  $\text{rk } E(K) = 0$ .

For  $u = 1$  this took most of a day, and a longer computation is running now to verify all of the cases.

If we pick a better basis, this works **integrally** and “recovers” Kummer’s proof, and generally gives lots of “divisibility” information.

E.g. for  $u = 1$ ,  $f_1(a, b, c)$  is a multiple of 7, and so  $z$  is too.

# Subgroups of $GL_2(\mathbb{Z}_2)$



# Agreeable groups

Zywina (indices occurring infinitely often, modulo conjectures)

The **index** of  $\rho_{E,N}(G_{\mathbb{Q}})$  divides 220, 336, 360, 504, 864, 1152, 1200, 1296 or 1536.

## Remark

$j(E) \in \{-7 \cdot 11^3, -7 \cdot 137^3 \cdot 2083^3\}$  *does occur finitely often*

## Definition

An open  $G \leq \mathrm{GL}_2(\widehat{\mathbb{Z}})$  is **agreeable** if  $\det G = \widehat{\mathbb{Z}}^\times$ , contains all scalar matrices, and the levels of  $G$  and  $G \cap \mathrm{SL}_2(\widehat{\mathbb{Z}})$  are divisible by the same odd prime divisors.

Zywina computes a finite list of agreeable  $G$  such that determining  $X_G(\mathbb{Q})$  determines all possibilities for the index.



## Agreeable groups

Zywina computes a finite list of agreeable  $G$  such that determining  $X_G(\mathbb{Q})$  determines all possibilities for the index.

Such  $G$  are **arithmetically maximal**, i.e.,  $X_G(\mathbb{Q})$  is finite and for every  $H$  such that  $G \leq H$ ,  $X_H(\mathbb{Q})$  is infinite.

If  $G$  is normal in  $H$  then the map of coarse spaces

$$X_G \rightarrow X_H$$

factors through a root stack  $\mathcal{X}$  over  $X_H$ .

$\mathcal{X}$  is a stacky  $\mathbb{P}^1$  or stacky elliptic curve, and is hyperbolic.

One can thus work with (twists of) étale covers of  $\mathcal{X}$  (e.g., Belyi covers) and mostly ignore  $G$ .

If  $G$  is not normal in  $H$ , one can look directly at the ramification and might get lucky.

# Root stacks

Let  $D \subseteq X$  be an effective divisor on a scheme and  $r \in \mathbb{Z}_{\geq 1}$ .

The  $r$ th root stack  $\sqrt[r]{(X, D)}$  of  $(X, D)$ :

$$\begin{array}{ccc} & \sqrt[r]{(X, D)} & \\ g \nearrow & \downarrow & \\ T & \xrightarrow{f} & X \end{array}$$

lifting  $f$  to  $g$  is the same as giving a divisor  $E$  on  $T$  and a linear equivalence  $rE \sim f^*D$ .